

TECHFORGOOD 2026

Internship Research Proposal

Project Title

NeuroAdapt: An AI-Powered Cognitive Accessibility Layer for Inclusive Web Navigation

Domain

Social Innovation & Inclusive Technology

Submitted By

TEAM DSY

Technology Stack

React.js • Node.js • MongoDB • AI APIs

Submitted To

IEEE MIT ADT Student Branch
IEEE Maharashtra Section

Date

May 2026

1. ABSTRACT

While modern web accessibility tools primarily focus on visual and auditory impairments, cognitive accessibility remains a massive global challenge. Over 15-20% of the global population is neurodivergent (including Dyslexia, ADHD, and Autism Spectrum), yet the modern internet is plagued with high text density, complex lexical jargon, and visually cluttered interfaces that induce cognitive overload. This project proposes NeuroAdapt, an AI-powered, full-stack cognitive accessibility ecosystem (Web Extension + Dashboard). Built on React.js and Node.js, NeuroAdapt leverages Natural Language Processing (NLP) APIs to dynamically restructure, simplify, and declutter web content in real-time. By bridging the digital divide for neurodivergent and elderly users, this solution champions inclusive technology, offering a highly scalable, software-based approach to digital equity.

2. PROBLEM STATEMENT

The Gap in Cognitive Web Accessibility

- What exact social issue exists? The internet is inherently hostile to individuals with cognitive processing difficulties. While screen readers help the visually impaired, there is a severe lack of tools that adapt the *comprehensibility* and *cognitive load* of digital content. High text density, complex vocabulary, and distracting UI elements prevent equitable access to digital education, e-governance, and online opportunities.
- **Who is affected?**
 - i. Neurodivergent Individuals: Specifically those with Dyslexia (affecting ~20% of the population) and ADHD.
 - ii. The Elderly: Who often face age-related cognitive decline and digital illiteracy.
 - iii. Non-Native Speakers & First-Generation Internet Users: Who struggle with complex institutional jargon (e.g., government portals, legal terms).
- Why current solutions fail? Current W3C Web Content Accessibility Guidelines (WCAG) predominantly address physical disabilities (e.g., color contrast, ARIA labels). Existing reading aids are static, requiring manual copy-pasting into third-party apps. They fail to provide *dynamic, in-place* contextual simplification.
- Why this problem matters now? As the world moves toward an entirely digital economy (education, banking, healthcare), digital exclusion based on cognitive ability is a modern human rights issue.

3. TENTATIVE PROJECT TITLE

NeuroAdapt: An AI-Powered Cognitive Accessibility Layer for Inclusive Web Navigation

4. PROPOSED SOLUTION

4.1 Project Overview

NeuroAdapt is a software-based accessibility ecosystem comprising a React.js Browser Extension and a Node.js/Express Backend. Instead of altering the underlying code of a website, it acts as an intelligent intermediary layer. When a user visits a complex website (e.g., a government portal or educational blog), the extension parses the DOM, sends the complex text to the Node.js backend, utilizes an AI API to simplify the jargon, applies Dyslexia-friendly formatting, and renders the adapted content seamlessly back into the user's browser.

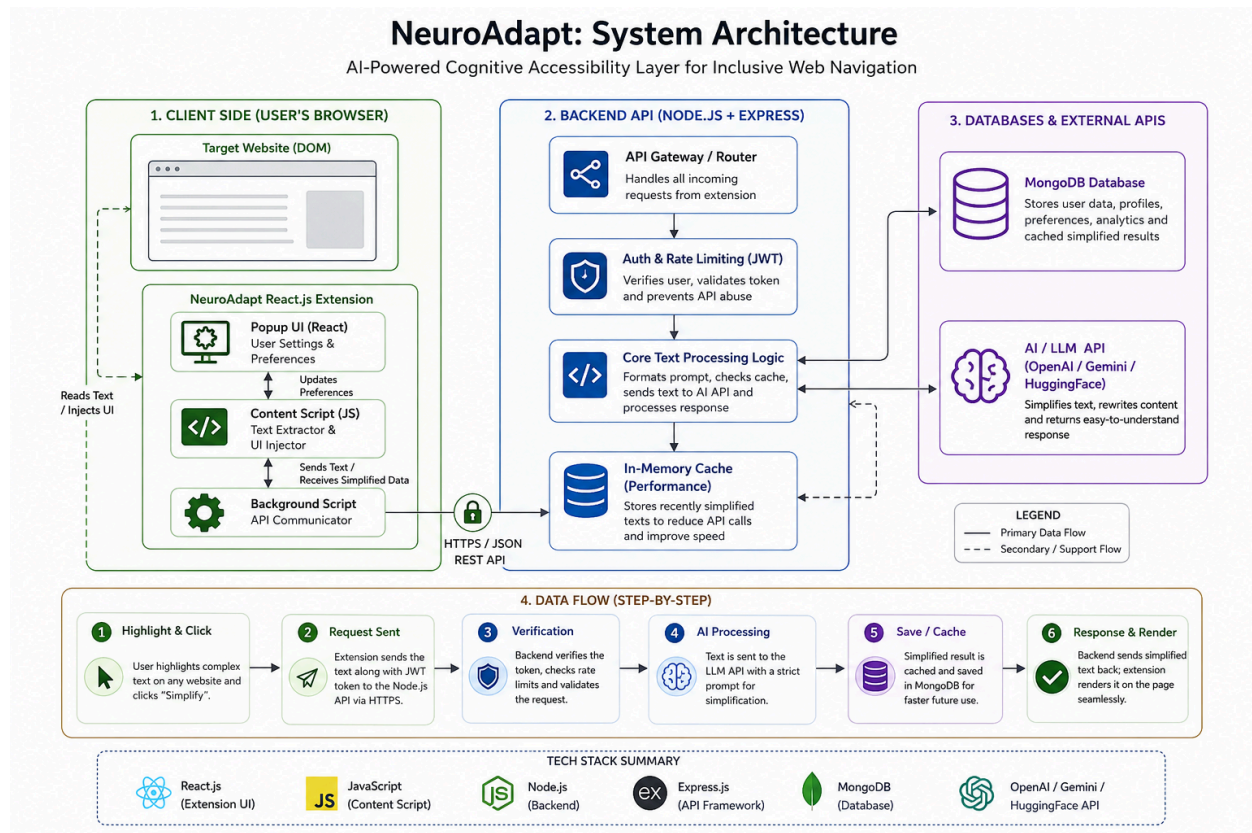
Target Users

- Students with learning disabilities (Dyslexia, ADHD).
- Elderly users navigating complex digital interfaces (healthcare, banking).
- Rural or first-generation digital users requiring simplified language.

Main Features (1-Month Feasibility)

1. Lexical Simplification (AI-Powered): Users can highlight complex paragraphs and click "Simplify." The Node.js backend uses an LLM API (e.g., OpenAI/Gemini) to rewrite the text into plain, easy-to-understand language.
2. Bionic / Focused Reading Mode: A JavaScript algorithm that dynamically bolds the first few letters of words to guide the eyes, proven to help users with ADHD maintain focus.
3. UI Decluttering (Focus Mode): Strips away distracting sidebars, ads, and pop-ups, rendering only the core article in a clean, high-contrast, customizable React UI.
4. Cloud-Synced Profiles: Users can log into the extension. Their preferences (font size, OpenDyslexic font toggle, color themes, simplification level) are stored in a MongoDB database and synced across devices.

4.2 System Architecture & Workflow



- 1. Client Layer (React.js Browser Extension):** Injects an accessibility widget into web pages, captures DOM content, monitors user interactions, and provides accessibility customization features such as text simplification and focus mode.
- 2. Content Processing Module:** Extracts webpage text, identifies complex content, and securely transfers user requests and webpage data to the backend server through REST APIs.
- 3. API Gateway & Backend Layer (Node.js/Express.js):** Handles request routing, JWT-based authentication, rate limiting, and communication between the browser extension, database, and AI processing services.
- 4. AI Processing Layer:** Integrates with external NLP/LLM APIs (OpenAI, Gemini, or HuggingFace) to perform lexical simplification, content summarization, and readability enhancement using optimized prompt engineering techniques.
- 5. Database Layer (MongoDB):** Stores user profiles, accessibility preferences, usage analytics, cached simplified content, and application configuration data for improved performance and personalization.

4.3 Suggested Tech Stack

- Frontend: React.js, Tailwind CSS (for rapid, accessible styling of the extension UI), Webpack/Vite.
- Backend: Node.js, Express.js, RESTful APIs.
- Database: MongoDB (Mongoose ORM) for flexible JSON document storage.
- AI Integration: OpenAI API (GPT-3.5/4-mini) or HuggingFace Inference API for text summarization.

4.4 Expected Outcomes (By end of 1 Month)

- A fully functional browser extension capable of modifying live web pages.
- A secure backend handling API keys and user data securely.
- A demonstrable prototype showing a "Before & After" of a complex website transformed into an accessible, neuro-inclusive format.

4.5 Why this is better than existing approaches

Unlike standard tools (like Grammarly) that help people *write*, NeuroAdapt helps marginalized people *read and comprehend*. Unlike standard accessibility tools that just read text aloud, NeuroAdapt fundamentally alters the *complexity* of the text, solving the root cause of cognitive overload.

5. FUTURE SCOPE & RESEARCH POTENTIAL



Research Publication Potential

Title Idea: *"Evaluating the Efficacy of LLM-Driven Lexical Simplification in Reducing Cognitive Load for Neurodivergent Web Users."* Research Focus: You can conduct a small user study during the internship. Measure how much faster/better users understand a Wikipedia page using NeuroAdapt vs. a standard browser. You can publish this empirical data at IEEE HCI (Human-Computer Interaction) or Accessibility conferences.



Scalability & Deployment Potential

This has immense B2B (Business-to-Business) and B2G (Business-to-Government) potential.

- B2B: Universities and schools can purchase enterprise licenses for their students to use on school devices.
- B2G: Governments can integrate this API directly into their e-governance portals to ensure WCAG 3.0 compliance.



Future Enhancements

- Image-to-Context (Vision AI): Generating descriptive, simplified alt-text for complex charts/graphs for visually and cognitively impaired users.
- Multilingual Cognitive Simplification: Translating complex English jargon into simplified regional languages (e.g., Hindi, Marathi) using tools like Bhashini API.
- Automated Content Warning: AI flagging content that might induce anxiety or sensory overload.

6. CONCLUSION

The NeuroAdapt project sits at the critical intersection of Software Engineering, Artificial Intelligence, and Social Inclusion. By building a scalable, software-only solution leveraging modern web technologies (React/Node) and AI, this project does not just conceptualize inclusivity—it actively engineers it. It meets all criteria for the TechForGood 2026 initiative by being achievable within a one-month timeframe, deeply aligned with the IEEE mission of "Advancing Technology for Humanity," and holding significant potential for academic publication and real-world deployment.